

Yalan Song (she/her)

Research Assistant Professor
yxs275@psu.edu; TEL: (814)826-5393;

The Pennsylvania State University
272 EcoRE Building, University Park, PA
Website: <https://yalan-song.github.io/>

ACADEMIC EXPERIENCE

2023-present	Research Assistant Professor	Pennsylvania State University
2022-2023	Postdoctoral Fellow	Pennsylvania State University

EDUCATION

2017-2022	PhD	Pennsylvania State University	Water Resources Engineering (Major)
2017-2022	PhD	Pennsylvania State University	Computational Science (Minor)
2014-2017	MS	Shanghai Jiao Tong University	Ocean Engineering
2010-2014	BS	Tianjin University	Ocean Engineering

FUNDING EXPERIENCE

- **External Funding**

- **Funded**

- * *Surface-subsurface Integrated Water Prediction System for Long-Term Assessment with Physics-Informed Differentiable Modeling*, funded by USGS (2024-2026), Role: Project co-PI, \$640,000
- * *Arctic Hydrologic and Thermal Dynamic Modeling*, LANL subaward (2025-2029), Role: Project co-PI, \$600,000
- * *Differentiable Hybrid Hydrologic Modeling for RFC Forecasting Operations*; invited for full proposal submission to NOAA in 2026 (final round); Role: Project co-PI, \$300,000 (2026 July-2027 March)
- * *CIROH Scoping Project: Large-domain Hydrologic Model Calibration Initiative*; invited for full proposal submission to NOAA in 2026 (final round); Role: Project co-PI, \$300,000 (2026 July-2027 March)

- **Pending**

- * *Taming the extremes: Improving operational flash flood forecasting in NextGen*; Submitted to NOAA in 2025; Role: Project lead PI
- * *Reimagining NextGen: Demonstrating the art of the possible in large-domain hydrologic prediction*; Submitted to NOAA in 2025; Role: Project co-PI
- * *Uncovering Causal Factors of Hydrological Drought to Inform Drought Early Warning System for Watersheds in Intermountain West*; Submitted to NOAA in 2025; Role: Project co-PI

- **Unfunded**

- * *Hydrology-informed Forcing Correction with Differentiable Models to Produce High Quality Historical and Forecast Forcing Datasets*; Submitted to NOAA in 2023; Role: Project lead PI
- * *National-scale high-resolution differentiable modeling for improved hydrologic forecasts and hydrology-informed forcing bias correction*; Submitted to NOAA in 2024; Role: Project lead PI
- * *Multivariate data learning and assimilation for ungauged basins in NextGen at hourly time scale*; Submitted to NOAA in 2024; Role: Project co-PI, \$1,000,000

- **Internal Funding**

- **Pending**

- * *A Coupled Plant-Soil-Water AI Framework for Predicting Coastal Agroecosystem Responses to Salinization and Marsh Invasion*; 2026 Huck Seed Grants; Role: Project co-PI
- * *Multimodal Deep Learning for Interpreting and Predicting Whole-Brain fMRI Dynamics*; 2026 Huck Seed Grants; Role: Project co-PI
- **Unfunded**
 - * *From Hydrology to Neutrinos and Back: Cross-Fertilizing Deep Learning Techniques*; Seed Grant in PSU; Role: Project co-PI

RESEARCH EXPERIENCE

- **Data-driven and hybrid Models in large-scale hydrology**
 - The [AI-based high-resolution differentiable \(physics-informed\) hydrologic modeling framework](#) for the next generation of Operational National Water Model.
 - An AI-based differentiable hydrologic model for improving [the prediction of unseen extreme flood events](#).
 - The [adjoint-based implicit solver](#) to enable advanced numerical schemes in the large-scale differentiable hydrologic modeling.
 - Long Short-Term Memory (LSTM) models in hydrology and environmental science for regional [snow water equivalent](#) and CONUS-scale [suspended sediment concentration](#).
- **Data-driven and hybrid models in computational hydraulics**
 - [AI-based surrogate models](#) for shallow water equations (SWEs) in computational hydraulics.
 - [River channel bathymetry inversion](#) using an AI-based surrogate model of SWEs.
 - [AI-based differentiable SWEs](#) for learning the distribution of key hydraulic parameters (roughness) in the river channel.
- **Classic computational environmental fluid dynamics**
 - [3D local scour \(hydrodynamic+sediment transport\) solver](#), *ibScourFoam*, for scour modeling around complex structures using OpenFOAM (C++).
 - 3D scour model for the remediation of [underwater unexploded ordnances](#)
 - [Review paper](#) about 3D scour modeling
 - Physically-based [sand slide method](#) for scour models.
- **Mentored research**
 - [Variational data assimilation](#) for AI-based differentiable hydrologic models to improve streamflow forecasts.
 - Using AI-based high-resolution differentiable hydrologic modeling framework for [global water assessment](#).
 - An ensemble of differentiable hydrologic models for a multi-model mosaic framework.

PUBLICATIONS

- First authorship papers
 - **Song, Y.**, Sawadekar, K., Frame, J.M., Pan, M., Clark, M.P., Knoben, W.J., Wood, A.W., Lawson, K.E., Patel, T. and Shen, C., 2026. Physics-informed, Differentiable hydrologic models for capturing unseen extreme events. *Water Resources Research*, 62(2), p.e2025WR040414.

- <https://doi.org/10.1029/2025WR040414>
- **Song, Y.**, Bindas, T., Shen, C., Ji, H.*, Knoben, W. J. M., Lonzarich, L.*, et al., 2025. High-resolution national-scale water modeling is enhanced by multiscale differentiable physics-informed machine learning. *Water Resources Research*, 61, e2024WR038928. <https://doi.org/10.1029/2024WR038928>
 - **Song, Y.**, Knoben, W.J., Clark, M.P., Feng, D., Lawson, K., Sawadekar, K.* and Shen, C., 2024. When ancient numerical demons meet physics-informed machine learning: adjoint-based gradients for implicit differentiable modeling. *Hydrology and Earth System Sciences*, 28(13), pp.3051-3077, <https://doi.org/10.5194/hess-28-3051-2024>.
 - **Song, Y.**, Chaemchuen, P., Rahmani, F., Zhi, W., Li, L., Liu, X., Boyer, E., Bindas, T., Lawson, K. and Shen, C., 2024. Deep learning insights into suspended sediment concentrations across the conterminous United States: Strengths and limitations. *Journal of Hydrology*, p.131573, <https://doi.org/10.1016/j.jhydrol.2024.131573>
 - **Song, Y.**, Tsai, W.P., Gluck, J., Rhoades, A., Zarzycki, C., McCrary, R., Lawson, K. and Shen, C., 2024. LSTM-based data integration to improve snow water equivalent prediction and diagnose error sources. *Journal of Hydrometeorology*, 25(1), pp.223-237, <https://doi.org/10.1175/JHM-D-22-0220.1>.
 - **Song, Y.**, Shen, C. and Liu, X., 2023. A surrogate model for shallow water equations solvers with deep learning. *Journal of Hydraulic Engineering*, 149(11), p.04023045, <https://doi.org/10.1061/JHEND8.HYENG-13190>.
 - **Song, Y.**, Xu, Y., Ismail, H. and Liu, X., 2022. Scour modeling based on immersed boundary method: A pathway to practical use of three-dimensional scour models. *Coastal Engineering*, 171, p.104037, <https://doi.org/10.1016/j.coastaleng.2021.104037>.
 - **Song, Y.**, Darzikolaei, S.A.M. and Liu, X., 2022. Scour around underwater unexploded ordnances (UXOs): An experimental and computational investigation. *Ocean Engineering*, 262, p.112146, <https://doi.org/10.1016/j.oceaneng.2022.112146>.
 - **Song, Y.**, Xu, Y., & Liu, X. (2020). Physically Based Sand Slide Method in Scour Models Based on Slope-Limited Diffusion. *Journal of Hydraulic Engineering*, 146(11), 04020074, [https://doi.org/10.1061/\(ASCE\)HY.1943-7900.0001814](https://doi.org/10.1061/(ASCE)HY.1943-7900.0001814).
 - **Song, Y.**, Lai, Y.G. and Liu, X., 2020. An improved immersed boundary method for simulating flow hydrodynamics in streams with complex terrains. *Water*, 12(8), p.2226. <https://doi.org/10.3390/w12082226>
 - Mentees (noted with *)' papers
 - Ji, H.*, **Song, Y.**, Bindas, T., Shen, C., Yang, Y., Pan, M., Liu, J., Rahmani, F., Abbas, A., Beck, H. and Lawson, K., 2025. Distinct hydrologic response patterns and trends worldwide revealed by physics-embedded learning. *Nature Communications*, 16(1), p.9169. <https://www.nature.com/articles/s41467-025-64367-1>
 - Jamaat, A.*, **Song, Y., corresponding author**, Rahmani, F., Liu, J., Lawson, K. and Shen, C., 2025. Update hydrological states or meteorological forcings? Comparing data assimilation methods for differentiable hydrologic models. *Journal of Hydrology*. <https://doi.org/10.1016/j.jhydrol.2025.134137>
 - Sawadekar, K.*, **Song, Y. corresponding author**, Pan, M., Beck, H., McCrary, R., Ullrich, P., Lawson, K. and Shen, C., 2025. Improving differentiable hydrologic

modeling with interpretable forcing fusion. *Journal of Hydrology*, p.133320. <https://doi.org/10.1016/j.jhydrol.2025.133320>

- Collaborative papers

- Clark, M.P., Knoben, W.J., Spieler, D., Grundemann, G.J., Thebault, C., Vasquez, N.A., Wood, A.W., **Song, Y.**, Shen, C., Carney, S. and van Werkhoven, K., 2026. Comment on: "Friends don't let friends use NSE or KGE for hydrologic model accuracy evaluation: A rant with data and suggestions for better practice". *Environmental Modelling & Software*, p.106869. <https://doi.org/10.1016/j.envsoft.2026.106869>
- Li, P., **Song, Y.**, Pan, M., Lawson, K. and Shen, C., 2025. Ensembling differentiable process-based and data-driven models with diverse meteorological forcing datasets to advance streamflow simulation. *Hydrology and Earth System Sciences*, 29(23), pp.6829-6861. <https://doi.org/10.5194/hess-29-6829-2025>
- Liu, X. and **Song, Y. corresponding author**, 2025. Scientific machine learning of flow resistance using universal shallow water equations with differentiable programming. *Water Resources Research*, 61(9), e2025WR040265. <https://doi.org/10.1029/2025WR040265>
- Knoben, W.J., Raman, A., Grundemann, G.J., Kumar, M., Pietroniro, A., Shen, C., **Song, Y.**, Thebault, C., van Werkhoven, K., Wood, A.W. and Clark, M.P., 2025. How many models do we need to simulate hydrologic processes across large geographical domains?. *Hydrology and Earth System Sciences*, 29(11), pp.2361-2375. <https://doi.org/10.5194/hess-29-2361-2025>
- Yang, Y., Pan, M., Feng, D., Xiao, M., Dixon, T., Hartman, R., Shen, C., **Song, Y.**, Sengupta, A., Delle Monache, L. and Ralph, F.M., 2025. Improving streamflow simulation through machine learning-powered data integration and its potential for forecasting in the Western US. *Hydrology and Earth System Sciences*, 29(20), pp.5453-5476. <https://doi.org/10.5194/hess-29-5453-2025>
- Liu, J., Shen, C., O'Donncha, F., **Song, Y.**, Zhi, W., Beck, H.E., Bindas, T., Kraabel, N. and Lawson, K., 2025. From RNNs to Transformers: benchmarking deep learning architectures for hydrologic prediction. *EGUsphere*, 2025, pp.1-21. <https://doi.org/10.5194/egusphere-2025-1706>
- Liu, X., **Song, Y.** and Shen, C., 2024. Bathymetry inversion using a deep-learning-based surrogate for shallow water equations solvers. *Water Resources Research*, 60(3), p.e2023WR035890. <https://doi.org/10.1029/2023WR035890>
- Shen, C., Appling, A. P., Gentine, P., Bandai, T., Gupta, H., Tartakovsky, A., ... & Lawson, K. 2023. Differentiable modelling to unify machine learning and physical models for geosciences. *Nature Reviews Earth & Environment*, 4(8), 552-567. <https://www.nature.com/articles/s43017-023-00450-9>
- Lai, Y.G., Liu, X., Bombardelli, F.A. and **Song, Y.**, 2022. Three-dimensional numerical modeling of local scour: A state-of-the-art review and perspective. *Journal of Hydraulic Engineering*, 148(11), p.03122002. [https://doi.org/10.1061/\(ASCE\)HY.1943-7900.0002019](https://doi.org/10.1061/(ASCE)HY.1943-7900.0002019)

PREPRINTS

- **Song, Y.**, Clark, M., Liu, J., Halgren, J., Lawson, K., and Shen, C. 2024. Prominent impacts of hydrologic scaling laws on climate risks. Preprinted and submitted to *Nature Water*, <https://doi.org/10.21203/rs.3.rs-4584048/v1>.

TEACHING EXPERIENCE

- Teaching assistant for Computing Methods and Water Resources Engineering.
- Guest instructor of AI for Geosciences (grad level) at Penn State and AI for Water (grad level) at Tsinghua University
- Instructor of the CIROH Developer Conference (University of Utah), the Knowledge-Guided Machine Learning (KGML) Workshop (University of Michigan), and an online Deep Learning for Water workshop

MENTORING EXPERIENCE

Mantee	Position	Publication	Other achievements
Kamlesh Sawadekar	Ph.D candidate	Journal of Hydrology	First place of poster at the 38th AMS
Amirmoez Jamaat	Ph.D candidate	Journal of Hydrology	-
Leo Lonzarich	Ph.D candidate	-	Outstanding Student Presentation at AGU 2024
Haoyu Ji	Ph.D candidate	Nature Communications	

SERVICE

- The mentoring committee in Department of Civil and Environmental Engineering at Penn State.
- Primary convener and co-convener of two sessions in AGU 2025 Annual Meeting.
- Reviewers of Water Resources Research, Hydrology and Earth System Sciences, AGU advances, Engineering Applications of Computational Fluid Mechanics, Advances in Water Resources, Journal of Geophysical Research - Earth Surface, Journal of Hydrology, Journal of Hydraulic Engineering, The Cryosphere, etc
- NSE proposal reviewer.

INVITED PRESENTATIONS

- Improving Operational Extreme Flood Forecasting and Climate Change Impact Assessment with Physics-Embedded Differentiable Modeling. EGU General Assembly 2026.
- Differentiable Physics-Informed Machine Learning Enhances High-resolution Hydrologic Modeling, AGU Annual Meeting 2025.
- High-resolution Differentiable Models for Operational National and Global Water Modeling Assessment, Plenary Keynote, CSDMS 2025 Annual Meeting.
- High-Resolution Differentiable Models for Operational National and Global Water Modeling and Assessment. EGU General Assembly 2025.
- Differentiable Modelling to Unify Machine Learning and Physical Models for Geosciences, AGU Chapman Remote Sensing of the Water Cycle: Sensors to Science to Society, Honolulu, HI, 2024.

AWARDS AND FELLOWSHIPS

2017	Outstanding graduates of Shanghai Jiaotong University, China
2014	American Bureau of Shipping Scholarship
2012	National Scholarship awarded by The Ministry of Education, China
2011-13	Honor Student of Tianjin University, China